

# Structural Ruptures in World-System Transitions

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## A Formal Account of Historical Change

### Abstract

Historical transformations of the world system are often described in terms of gradual evolution, economic cycles, or technological progress. However, many pivotal transitions appear discontinuous: the existing institutional and economic structures cease to be valid descriptions of the system, and a new structural regime emerges. This paper proposes a formal framework for analyzing such transitions using the concept of **structural rupture**.

A world system is modeled as a configuration space equipped with a structural constraint operator  $S$  determining admissible configurations. Historical change becomes structurally significant when configurations previously admissible under  $S$  cease to be valid, requiring a new operator  $S'$ . We show that many major transformations in global history—from the collapse of medieval fortification systems after the introduction of artillery to the emergence of the Atlantic world economy—can be understood as **non-derivable structural transitions**  $S \rightarrow S'$ .

This formalization clarifies why transformative historical shifts are rarely predictable from within the prior system: the rupture invalidates the structural conditions that previously defined admissibility. The framework provides a bridge between historical analysis and formal systems theory, offering a new way to model long-term structural change in human societies.

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## 1 Introduction

Historians of economic and political systems have long observed that global history unfolds through a series of large-scale transformations. The transition from the Mediterranean-centered economy of the late medieval world to the Atlantic system of early modern Europe is one such example. Others include the Industrial Revolution, the emergence of global financial capitalism, and the contemporary digital transformation.

Traditional explanations describe these transitions in terms of technological innovation, shifts in trade networks, or changes in political power. While such explanations identify important causal factors, they rarely clarify **why the existing system suddenly becomes inadequate to describe reality**.

This paper proposes a formal perspective: many historical transitions are best understood as **structural ruptures**. In such cases, the underlying structure defining admissible configurations of the system ceases to hold, requiring the emergence of a new structure.

The central claim is that transformative historical changes are not merely dynamic developments within a system, but **transitions between structurally distinct regimes**.

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## 2 Historical Background

Several intellectual traditions have studied large-scale historical transitions.

World-systems analysis, associated with scholars such as **Immanuel Wallerstein**, interprets global history as a succession of economic systems characterized by changing cores, peripheries, and trade networks. Similarly, historical analyses of hegemonic cycles have described the succession of dominant powers in the global economy.

Economic theory has also addressed structural transformation through the concept of **creative destruction**, most famously developed by **Joseph Schumpeter**. In this view, technological innovation periodically disrupts established economic structures, allowing new industries and institutions to emerge.

However, these approaches remain largely descriptive. They identify patterns of transformation but do not provide a formal framework for understanding the structural conditions under which a system becomes invalid.

The approach developed here seeks to formalize these transitions using the language of structural operators and admissibility conditions.

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## 3 Structural Framework

Consider a world system represented by a configuration space

$$\Omega$$

whose elements represent possible configurations of institutions, technologies, economic relations, and political structures.

A structural constraint operator

$$S : \Omega \rightarrow \{0, 1\}$$

determines which configurations are admissible within the system.

Configurations satisfying

$$S(\omega) = 1$$

are structurally compatible with the system, while configurations for which

$$S(\omega) = 0$$

are incompatible.

Within a stable historical regime, system dynamics operate inside the admissible domain defined by  $S$ . Let

$$X = \{\omega \in \Omega \mid S(\omega) = 1\}$$

be the set of admissible states.

Historical processes such as trade expansion, political competition, or technological improvement correspond to trajectories within this admissible space.

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## 4 Structural Rupture

A **structural rupture** occurs when the existing structural operator  $S$  no longer provides a valid description of the system.

Formally, there exists a time  $t^*$  such that

$$S(\omega_{t^*}) = 0$$

for configurations that arise within the evolving system.

This does not necessarily imply physical impossibility. Rather, it indicates that the existing structural framework is no longer sufficient to describe the system.

A new structural operator must then be introduced:

$$S' : \Omega' \rightarrow \{0, 1\}$$

with a corresponding configuration space  $\Omega'$ .

The transition

$$S \rightarrow S'$$

constitutes a structural transformation of the system.

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## 5 Non-Derivability of Historical Transitions

An important implication of structural rupture is **non-derivability**.

Within the original system, dynamics are governed by some evolution operator

$$F : X \rightarrow X$$

which maps admissible states to admissible states.

However, if a configuration satisfying

$$S(\omega) = 0$$

is encountered, the dynamic evolution defined by  $F$  cannot proceed.

The trajectory reaches the boundary of admissibility.

As a result, the transition to the new structural regime  $S'$  cannot be derived internally from the dynamics of the original system.

This explains why major historical transformations are rarely predictable from within the conceptual framework of the existing system.

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## 6 Case Study: From Mediterranean to Atlantic System

One example of structural rupture can be found in the transition from the late medieval Mediterranean world to the early modern Atlantic system.

For centuries, Mediterranean trade networks and fortified cities formed the backbone of European economic and political organization. The structural operator  $S$  defining this system included assumptions about maritime routes, military technology, and trade geography.

The development of large-scale artillery fundamentally altered this structure. Fortifications designed for earlier forms of warfare became vulnerable to bombardment.

The fall of Constantinople in 1453 symbolized the collapse of a structural regime that had governed Mediterranean geopolitics for centuries.

At the same time, maritime technologies such as the ocean-going sailing ship enabled the exploration of new trade routes. The resulting discovery and colonization of the Americas transformed the geographical scope of economic exchange.

The system shifted from a Mediterranean-centered structure to an Atlantic-centered one.

In formal terms, the historical transition required a new structural operator  $S'$  describing the emerging global economic system.

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## 7 Innovation and Structural Stress

Technological innovations often act as **structural stress tests**.

An innovation does not automatically produce a systemic rupture. Instead, it introduces configurations that may or may not remain admissible under the existing structure.

If these configurations remain compatible with the operator  $S$ , the system adapts without fundamental transformation.

However, when innovations generate configurations for which

$$S(\omega) = 0$$

the system faces structural tension. If the new configurations prove economically or politically dominant, the existing structure collapses and a new one emerges.

Thus, innovation acts not merely as a dynamic driver but as a **test of structural validity**.

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## 8 Conclusion

Historical transformations are often interpreted as gradual evolutionary processes. However, many pivotal moments in world history appear better described as structural ruptures.

By modeling world systems as configuration spaces governed by structural constraint operators, it becomes possible to formalize these transformations.

Within this framework, historical change occurs not only through internal dynamics but also through transitions between distinct structural regimes.

Technological innovations, geopolitical shifts, and economic transformations may introduce configurations that invalidate the existing structure of the system. When this occurs, the system undergoes a structural transition  $S \rightarrow S'$ .

This perspective provides a new conceptual bridge between historical analysis and formal systems theory. It suggests that many large-scale transformations in human history—from the emergence of global trade networks to the digital revolution—can be understood as **non-derivable structural transitions in complex social systems**.

Medieval system (N0)



Artillery rupture (N1)



Atlantic system (N0')



Industrial rupture (N1')



Industrial world system (N0'')



Digital rupture (N1''')

Table — Structural Ruptures in World-System Transitions

| INNOVATION          | PREVIOUS SYSTEM               | STRUCTURAL STRESS                   | STRUCTURAL TRANSITION                 | NON-DERIVABILITY INSIDE THE PREVIOUS SYSTEM                                                   | HISTORICAL CONSEQUENCE                 |
|---------------------|-------------------------------|-------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------|
| Heavy plow          | Shallow-soil agriculture      | Deep soils become cultivable        | Expansion of agricultural production  | Existing agrarian techniques could not exploit heavy soils within the previous farming system | Medieval population expansion          |
| Printing press      | Manuscript knowledge economy  | Mass text reproduction              | Decentralized knowledge dissemination | Scribal production could not scale information transmission                                   | Reformation and scientific revolution  |
| Gunpowder artillery | Fortified medieval polities   | Walls lose defensive validity       | Territorial states dominate warfare   | Fortress defense assumptions collapse                                                         | Decline of feudal military structures  |
| Oceanic navigation  | Mediterranean-centered trade  | New oceanic routes viable           | Atlantic world-system emerges         | Land–Mediterranean trade networks could not incorporate transoceanic logistics                | European maritime expansion            |
| Steam engine        | Craft and agrarian production | Mechanical energy becomes scalable  | Industrial production system          | Manual and animal energy cannot scale productivity                                            | Industrial Revolution                  |
| Mechanical reaper   | Labor-intensive agriculture   | Drastic labor productivity increase | Mechanized agricultural production    | Traditional agrarian labor equilibrium becomes impossible                                     | Rural depopulation and urban migration |

| INNOVATION         | PREVIOUS SYSTEM $S$              | STRUCTURAL STRESS                              | STRUCTURAL TRANSITION $S \rightarrow S'$ | NON-DERIVABILITY INSIDE THE PREVIOUS SYSTEM                                          | HISTORICAL CONSEQUENCE           |
|--------------------|----------------------------------|------------------------------------------------|------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------|
| Railways           | Localized regional markets       | Transport costs collapse                       | Integrated national economies            | Pre-industrial transport systems cannot support large-scale industrial distribution  | National industrial markets      |
| Electricity        | Central mechanical power         | Energy becomes spatially flexible              | Electrified industrial systems           | Mechanical shaft systems cannot distribute power across large factories              | Modern industrial infrastructure |
| Nuclear technology | Conventional warfare             | Destructive power exceeds political structures | Nuclear deterrence regime                | Classical military strategy cannot operate under mutually assured destruction        | Cold War strategic balance       |
| Internet           | Hierarchical information systems | Instant global connectivity                    | Networked digital society                | Industrial-era information hierarchies cannot process real-time global communication |                                  |

*A particularly clear example of structural rupture occurs in agriculture. The introduction of mechanized harvesting dramatically reduced labor requirements. Tasks that previously required ten workers could be performed by a single operator. The resulting rural depopulation was not primarily a political phenomenon but a structural consequence of technological change.*



# Additional Structural Phase Transitions of Civilization

| INNOVATION / TRANSITION         | PREVIOUS SYSTEM $S$          | STRUCTURAL STRESS                         | STRUCTURAL TRANSITION $S \rightarrow S'$ | NON-DERIVABILITY INSIDE PREVIOUS SYSTEM                                  | HISTORICAL CONSEQUENCE                               |
|---------------------------------|------------------------------|-------------------------------------------|------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------|
| <b>Coined money</b>             | Tribute and barter economies | Exchange becomes abstract and portable    | Monetary economic system                 | Barter networks cannot scale to large anonymous markets                  | Emergence of long-distance trade and early markets   |
| <b>Double-entry bookkeeping</b> | Informal merchant accounting | Capital flows become complex              | Financial capitalism                     | Traditional accounting cannot represent multi-partner capital structures | Rise of modern banking and large commercial firms    |
| <b>Scientific method</b>        | Authority-based knowledge    | Empirical verification replaces tradition | Self-correcting scientific system        | Scholastic reasoning cannot generate cumulative experimental knowledge   | Scientific revolution and technological acceleration |

# Comparative Table

*Although structural transitions in world-systems may superficially resemble Marxist analyses of historical change, the present framework differs fundamentally. In Marxist theory, transformation emerges from social contradictions within a mode of production. In contrast, the model proposed here defines rupture formally as the appearance of configurations that cannot be admitted within the structural operator governing the previous system. Historical transitions therefore arise from structural non-derivability rather than class conflict or teleological development.*

| ASPECT                              | MARXISM                               | TNA MODEL                                                                              |
|-------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------|
| Driving mechanism of change         | Class struggle                        | Structural incompatibility                                                             |
| Mode of explanation                 | Socioeconomic                         | Formal / systemic                                                                      |
| Historical trajectory               | Teleological                          | Open-ended                                                                             |
| Role of technology                  | Subordinate to social relations       | Possible trigger of structural rupture                                                 |
| Predictability of historical change | History partially predictable         | Structural ruptures are non-derivable                                                  |
| Nature of systemic transition       | Transformation of modes of production | Structural rupture ( $S \rightarrow S'$ )                                              |
| Formal representation of transition | Change in mode of production          | Structural transformation ( $S \rightarrow S'$ ) generating a new admissibility regime |